

JAN 2027 ① START DRAFTING JOURNAL ARTICLE FOR VALIDATION RESULTS →	FEB 2027	MAR 2027
APR 2027 ① MAKE CONFERENCE SLIDES ↓ ⑦ ⑮ M+C CONFERENCE ~ COLUMBUS, OH ↓ ②②	MAY 2027 ① TARGET PRELIM EXAM! ⑦ START LUN RESEARCH	JUN 2027
JUL 2027	AUG 2027	SEP 2027 ① RESTART NSF FUNDING ③① FINAL DELIVERABLES DUE FOR LANL
OCT 2027	NOV 2027	DEC 2027

★ Ask Najel why "failure cannot occur in the control drum."

reactivity $\rightarrow \rho = \frac{\Delta K}{K} \left\{ \begin{array}{l} \% \text{ change} \\ \text{in criticality } K \end{array} \right.$

$\rightarrow$ study tried 8, 4, 2, & 1 drum cases

Control methods

PID: proportional integral derivative control

SMC: sliding model control

NMPC: nonlinear model predictive control

★ What does the $\leq 1100K$ fuel temp. constraint mean?

★ Should we try to validate all physical control algorithms? Or just best performing?

SPECIFICATION

Load-following simulation

ENVIRONMENT

Holos-Quad

HTGR Reactor

- nonlinear point kinetics
- thermal hydraulics
- poison + reactivity

Limitations:

- no spatial (only temporal) effects

* assumes uniform neutron distribution throughout the core

need a high fidelity model to test this on!
LBRELAP or SAM

SYSTEM

CONTROL ALGORITHM

(PID, STC, NMPC, RL, MARL)

GOAL:

validate this control algorithm!

Failure is defined as $\geq 1\%$ deviation from target power. (e.g.)

AND

available action is drum speeds (individual) as $\frac{d\theta}{dt}$, $dt = 1$ second (fixed)

★ do we force all CDs to operate symmetrically?

★ is the answer true for MARL & RL?

Fuel Temp.
Mod. Temp.
Coolant Temp.
Control Drum rotation
Control Drum Speed

Operating Variables

NEED TO CREATE SPECIFICATIONS FOR ALL OF THESE VARIABLES!

Core reactivity
Fuel temp.
Mod. temp.
Coolant temp.

Plant Variables

PER-STATE OBSERVATIONS

$P_t \equiv$ current measured power

$P_{t+1}^* \equiv$ the next desired power

$P_{t-1} \equiv$ the previous measured power

$\theta_t^1, \dots, \theta_t^8 \equiv$ the current control drum positions

*Noise was added in Leo's paper to evaluate the impact.

↳ we will conduct a much more intensive version of this

→ 2 million timesteps!

*why would you let the 8 control drums be perturbed separately if the model assumes a spatially uniform neutron distribution?

★ MARL $\frac{1}{2}$ RL outperform PID when noise is added

"Additional work...exploring the impact of biased measurements would be an important next step"